

# PHLOTON

Reliable cold chain where the grid ends: for outreach clinics, blood banks, labs and the last child on the route.

THE DEVICE, IN THREE WORDS

## Active. Off-grid. Connected.

A solar-charged, connected carrier that holds vaccines, blood and biologics at 2–8 °C where the grid ends. Fifty units in service in India; built for the world.

Prerit Mittal, founder • [prerit@phlton.com](mailto:prerit@phlton.com) • [phlton.com](http://phlton.com)



2 L units in assembly, Bengaluru

# Where the grid ends, **the medicine stops.**

About a billion people rely on health facilities without reliable power. Their vaccines, insulin and blood travel the last kilometres in ice boxes that need a freezer upstream and record nothing. Phlodon is an active, solar-charged, connected carrier built for exactly that leg.

**13.5<sup>M</sup>**

children received no vaccine at all in 2025

WHO / UNICEF, July 2026

**1<sup>B</sup>**

people in lower-income countries served by health facilities without reliable power

WHO • World Bank • IRENA, 2023

**50%**

of vaccines wasted every year at WHO's upper estimate; up to a quarter lost to temperature alone

WHO, via UNEP 2020 • IQVIA 2025

## WHAT ONE UNIT DOES

On our model a unit protects about **1,900 doses a year**: 7,500 doses handled on outreach × 25% baseline wastage. Over a five-year life that is 9,500 doses for a \$1,000 lifetime cost — about **11 cents a dose**, before the blood bags and samples it also carries.



## WHAT THE FLEET DOES

**43 million doses a year by FY30** across 22,800 units, and 10,200 t CO<sub>2</sub>e avoided a year: no compressor, no refrigerant, charged by the sun. Company plan, August 2026.

# Half of what the world makes never protects anyone.

By WHO's upper estimate, up to half of all vaccines are wasted every year; IQVIA puts the loss to temperature alone at 7–25%. About a billion people in lower-income countries depend on health facilities without reliable power. A cold chain that needs a grid and a freezer cannot follow the medicine to the patient.

50%

of vaccines wasted every year at WHO's upper estimate; up to a quarter lost to temperature alone

WHO, via UNEP 2020 • IQVIA 2025

\$35<sup>B</sup>

lost by biopharma yearly to temperature failures

IQVIA Institute estimate, 2019

1<sup>B</sup>

people in lower-income countries served by health facilities without reliable power

WHO • World Bank • IRENA, 2023

## HOW THE LAST MILE IS SOLVED TODAY

Ice-pack and phase-change-material (PCM) carriers, conditioned in a freezer upstream — or a mains-charged active carrier at about \$2,000.

## WHY THAT FALLS SHORT

A passive carrier needs a working freezer, loses about half of what is left each time it is opened (our field rule of thumb), can freeze the payload, and records nothing. A mains-charged active carrier has to come back to a socket to recharge; the grid sets its range.

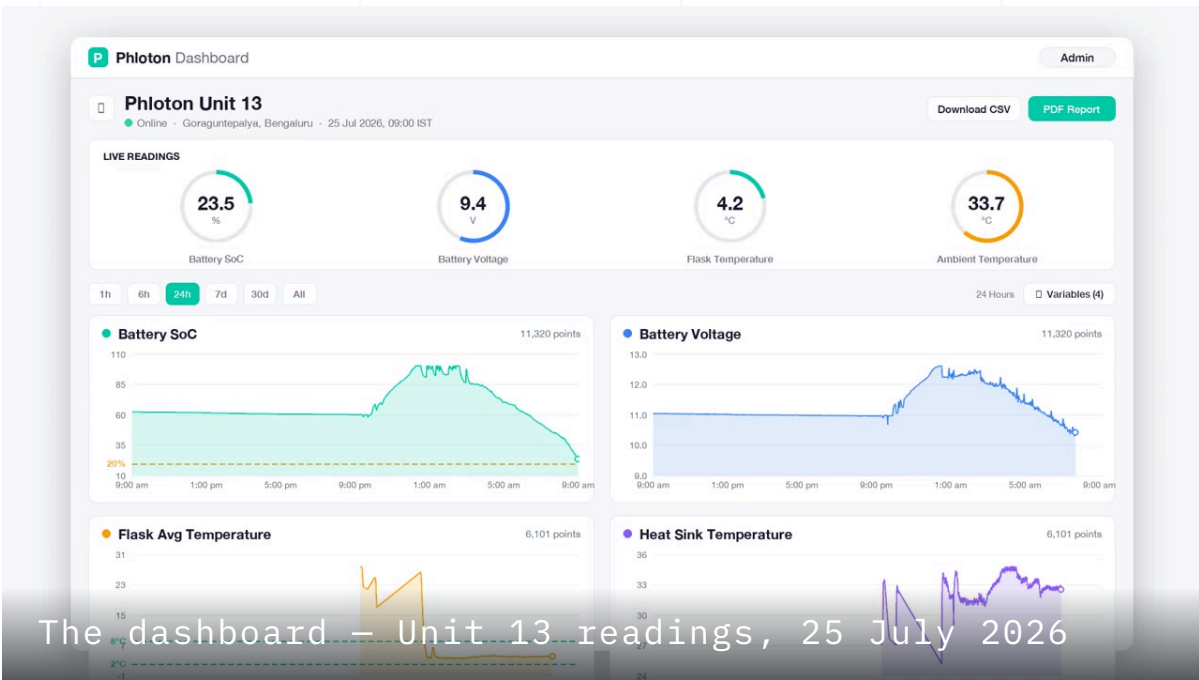
# 2–8 °C. Designed for sixteen hours at 43 °C. Charged by the sun.

A 3.75 kg, 2-litre device with a solid-state cooling engine: no ice, no freezer, no refrigerant. It holds 2–8 °C, charges from the sun, a vehicle or the wall, and reports every reading over 4G. From a warm start an empty chamber is inside 2–8 °C in fifteen to twenty minutes; the status light turns green when the chamber is in band.

|                  |                  |               |             |
|------------------|------------------|---------------|-------------|
| AMBIENT • DESIGN | PAYLOAD • TARGET | HOLD • DESIGN | AUDIT TRAIL |
| 43 °C            | 3 °C             | 16 h          | Every trip  |

### WHAT THE CUSTOMER SEES

Sixteen sensing points reported over 4G, with an offline buffer for the stretches without signal: each unit's readings and temperature trace, alerts when a reading drifts, the fleet on a map, and a compliance report at the end of every trip.



# One engine. Every payload that must stay cold.

Every proven row below runs on the same solid-state cooling engine and controller. A new payload is a new customer, not a new engine; the larger sizes add a coolant loop.

| PAYLOAD                                          | HOLD BAND              | WHERE IT STANDS                                                                                                                     |
|--------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| PROVEN IN THE FIELD                              |                        |                                                                                                                                     |
| Vaccines                                         | 2–8 °C                 | SELCO Foundation outreach clinics · PCM jackets for open-vial sessions                                                              |
| Blood bags                                       | 2–6 °C · never freezes | Raktamitra blood bank, Government Institute of Medical Sciences (GIMS), Noida — transported with no haemolysis (no red-cell damage) |
| Diagnostic samples                               | 2–8 °C                 | Beckman Coulter — six months of 7–8 h runs, then a purchase                                                                         |
| Specialty biologics                              | 2–8 °C                 | Roche India, Kolkata — cold-chain-as-a-service, depot to hospital · starting                                                        |
| Veterinary & dairy                               | 2–8 °C                 | Akshayakalpa · URMUL                                                                                                                |
| NEXT — SAME ENGINE, NEW SET POINT OR FORM FACTOR |                        |                                                                                                                                     |
| Insulin & antivenom                              | 2–8 °C                 | 2.5 L for primary health centres                                                                                                    |
| Reagents, enzymes, trial kits                    | programmable           | 5 L for labs and clinical-trial logistics                                                                                           |
| Organs & tissue                                  | 4–10 °C                | 10 L · exploratory, with a transplant partner                                                                                       |

# Active, off-grid, connected. Only one does all three.

The last mile needs three things: active cooling, power without a grid, and a live record. The table scores each option against all three.

|                          | ICE - PACK CARRIER             | PCM CARRIER                            | BLACKFROG EMVOLIO       | PHLOTON                                                                 |
|--------------------------|--------------------------------|----------------------------------------|-------------------------|-------------------------------------------------------------------------|
| Price                    | ~\$75 prequalified + a freezer | varies + a freezer                     | ~\$2,000                | \$700 → \$500 at scale                                                  |
| Holdover                 | set by the ice packs           | fixed window, halves with each opening | 12 h, verified at 43 °C | 16 h at 43 °C · 24 h at 27 °C (design)                                  |
| Weight · battery         | —                              | —                                      | 5.5 kg · 200 Wh         | 3.75 kg · 390 Wh                                                        |
| Charging                 | needs a freezer                | needs a freezer                        | mains                   | solar, vehicle or mains                                                 |
| Monitoring               | none                           | none                                   | live tracking           | 16 sensing points · 4G with offline buffer · per-trip compliance report |
| WHO PQS prequalification | listed (E004)                  | varies                                 | listed (E003)           | on the E003 route — Emvolio's category — funded by this round           |

WHO PQS is the prequalification UN agencies and immunisation programmes buy against: E003 covers powered cooling devices, E004 insulated containers  
Patent filed · CDSCO (India's medical-device regulator) test licence · no refrigerant at all · five-year cost of ownership about a third below passive, on our model

# \$700 device. \$60 a year. 5-year warranty.

Two revenue lines — the device, with a five-year warranty, and a service plan — and, for institutions, the whole thing as a contract.

| LINE                    | PRICE                         | WHAT'S INCLUDED                                                                                                                                                 |
|-------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device • 2 L            | \$700 today<br>\$500 at scale | Device, charger and five-year warranty. Sold near cost at pilot volumes; at scale, \$500 against a build of about \$200 after enclosures are injection-moulded. |
| Service plan            | \$60 per device per year      | 4G telemetry, alerts, per-trip compliance reports, maintenance. About 14% of revenue by year five.                                                              |
| Cold-chain-as-a-service | Priced per contract           | We carry the payload and own the audit trail; they own the patient. First engagement starting with Roche India, Kolkata: depot to hospital.                     |

\$1,000

Lifetime value per device over five years: \$700 device + 5 × \$60 service.

60%

Gross margin at scale — \$500 price against a ~\$200 build. Near cost today.

B2B

For three years — pharma, diagnostics chains, labs and healthcare institutions, reached through pharma distributors and hospital groups. Consumer comes later, by choice.

# Fifty units in service.

## All fifty bought.

Forty bought commercially, ten through grant programmes. Beckman Coulter ran diagnostic samples on 7–8 hour routes for six months, reported no excursions and no sample loss — then bought.

96%

of 1,883 monitored hold-hours in 2–8 °C – chamber telemetry from 17 connected units, January–August 2026

0

readings below 0 °C, fleet-wide – nothing freezes

0

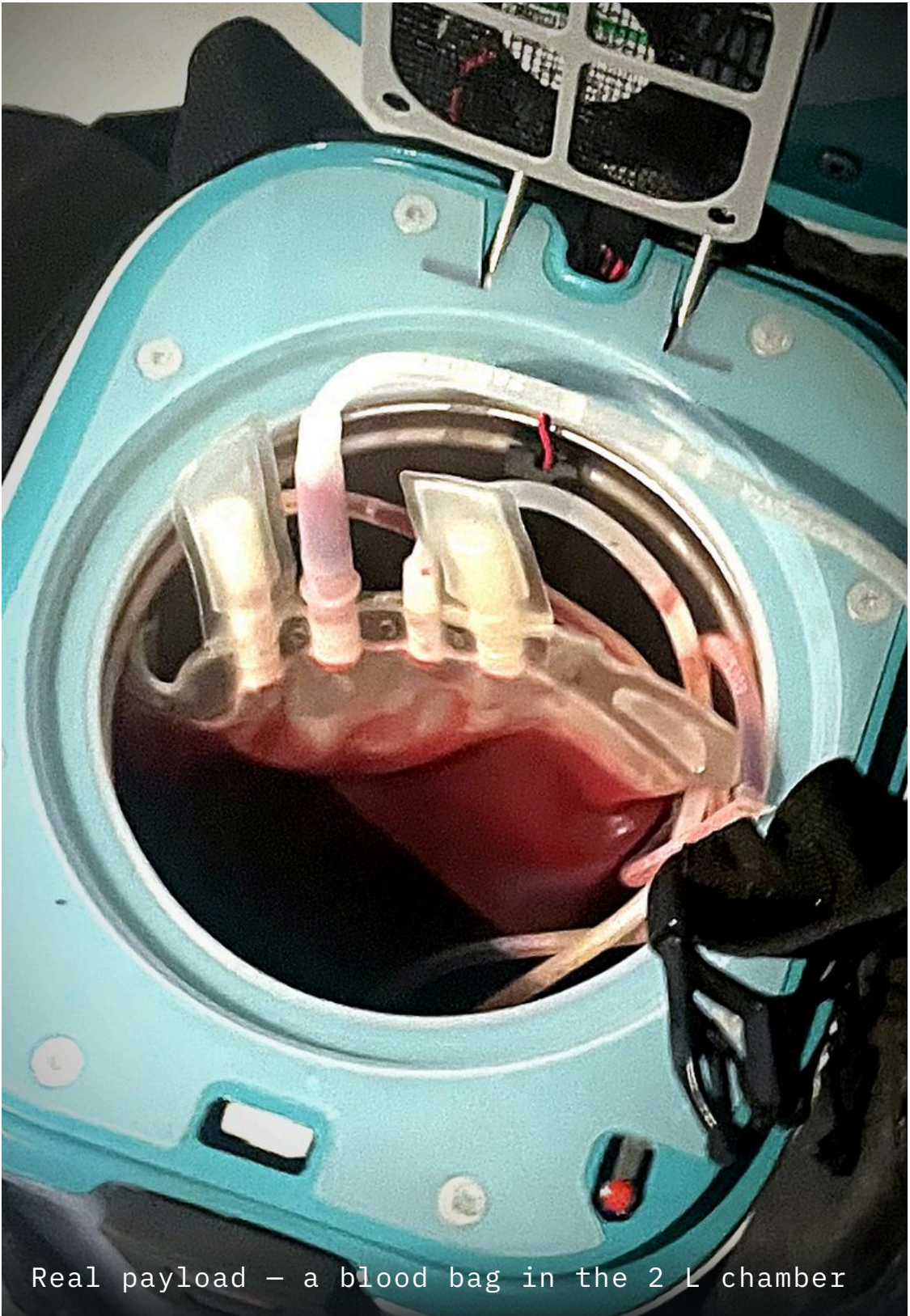
excursions reported in six months of 7–8 h sample runs at Beckman Coulter

IN HAND

CDSCO (India's medical-device regulator) test licence · patent filed · EMI/EMC testing at TÜV SÜD to IEC 61326-1 · IP55 dust-and-water ingress, tested in a NABL-accredited lab · founder trained as ISO 13485 internal auditor

NEXT 12 MO

ISO 13485 quality system + CDSCO manufacturing licence (Form MD-5) · WHO PQS E003 submission



Real payload – a blood bag in the 2 L chamber

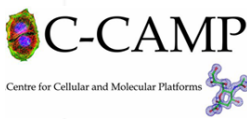
# Who buys it. Who backs it.

Beckman Coulter, Roche India and Solumatix are the three commercial anchors (pages 5 and 8). Around them, clinics, hospitals, dairy and research — with units reporting from five Indian states.

## CUSTOMERS, PILOTS & DEPLOYMENTS

|                                                                                                                            |                                                                                                                                    |                                                                                                                                                                  |                                                                                                                                                  |                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <br>Diagnostics • bought after six months | <br>Cold-chain-as-a-service • Kolkata • starting |  <b>SELCO</b> Foundation<br>Primary-health outreach                           |  <b>Swami Vivekananda Youth Movement</b><br>District hospital | <br>Dairy & veterinary                |
| <br>Eye care                             | <br>Rural health, Rajasthan                     |  <b>TIGS</b><br>Tata Institute for Genetics and Society<br>Genetics research | <br>Screening                                                | <br>30-unit order • advance received |

## INCUBATED & FUNDED BY

|                                                                                                  |                                                                                                       |                                                                                                                           |                                                                                                           |                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>SEED fund | <br>TIDE 2.0      | <br>NIDHI PRAYAS                     | <br>Incubation       | <br>Incubation                                                     |
| <br>Grant     | <br>IQ 2.0 grant | <br>Leaders in Innovation Fellowship | <br>DPIIT recognised | <br>Karnataka • Tamil Nadu • Maharashtra • Gujarat • Uttar Pradesh |

# We need one percent of this market.

\$1.2 billion addressable; \$0.7 billion in the regions we can reach, the Gulf included. One percent is \$12 million a year by year five, and our plan reaches two-thirds of it — B2B for three years.

\$1.2<sup>B</sup>

**TAM** — about 1 million last-mile cold-chain devices in use worldwide × ~\$1,200 average lifetime value at market prices. Company bottom-up estimate.

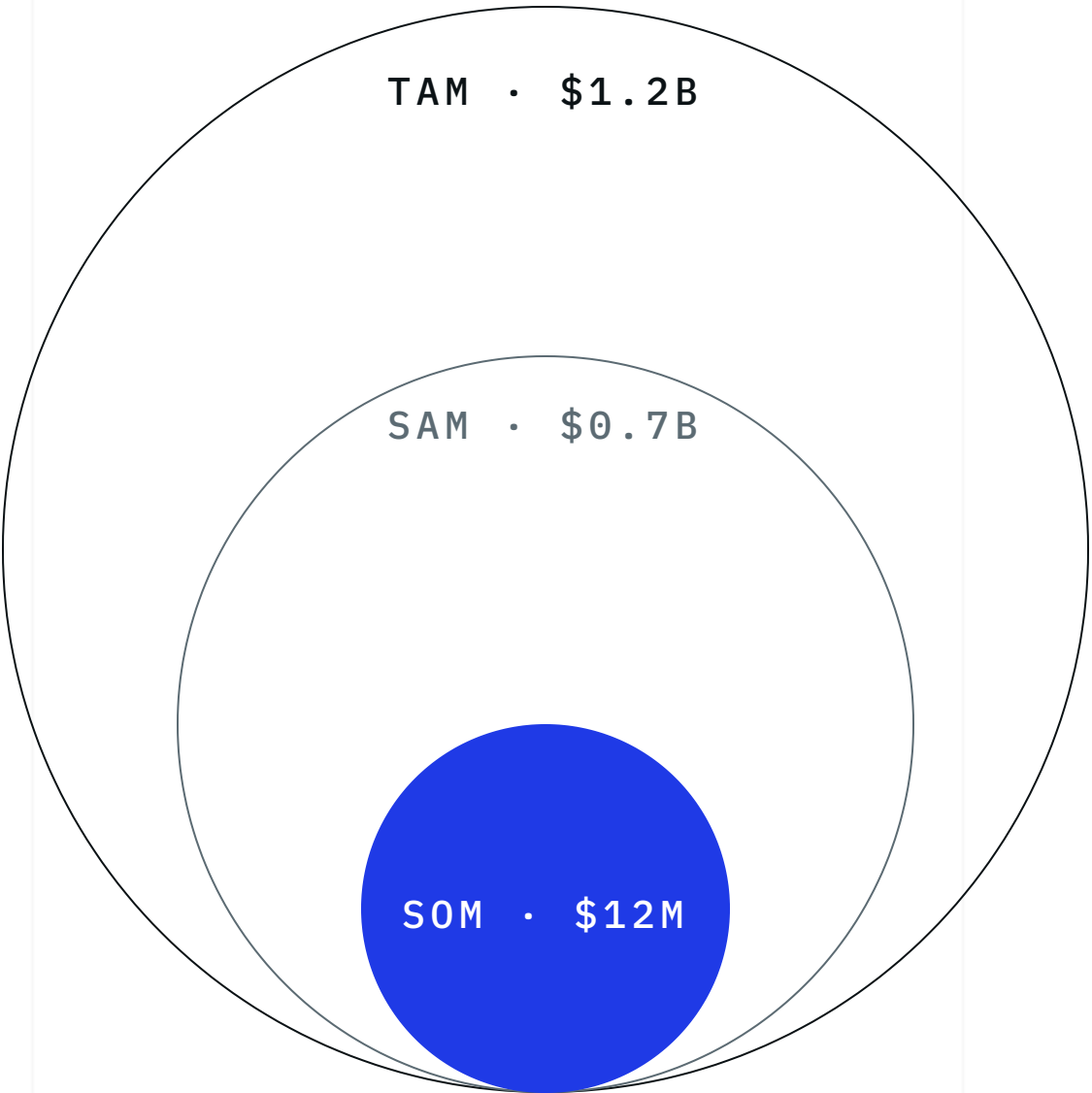
\$0.7<sup>B</sup>

**SAM** — India, the Middle East, Central Asia, Africa and Southeast Asia: the geographies our distributor model can reach.

\$12<sup>M</sup>

**SOM** — 1% of TAM a year by year five, deliberately modest. Our year-five plan of \$7.9M sits inside it.

Context: pharma spends \$21.3B a year on cold-chain logistics (IQVIA, 2025). In Kazakhstan and Uzbekistan about 9 in 10 medicines are imported, so cold-chain products travel a long way to the patient — which is why Central Asia sits inside the SAM. TAM is our separate bottom-up estimate of the last-mile device market.



# Made anywhere. Serviced by local technicians.

It is solid-state — no compressor line, no refrigerant to handle — so it can be built in India today and assembled in any region when volume or local rules call for it. Local technicians are trained with every deployment, and 4G diagnostics tell them what to fix before they travel. The need is wherever the grid is not (top row); on our model, this is what each unit adds (bottom row).

3.2<sup>B</sup>

vaccine doses UNICEF delivered to 103 countries in 2025 – all of them cross a last mile

UNICEF Annual Results, 2025

13.5<sup>M</sup>

children received no vaccine at all in 2025

WHO / UNICEF, July 2026

1<sup>in 2</sup>

hospitals in sub-Saharan Africa lack reliable electricity; 15% of facilities have none

WHO • World Bank • IRENA, 2023

1,900

doses protected per unit, per year – 7,500 doses handled on outreach × 25% baseline wastage

Company model

43<sup>M</sup>

doses a year by FY30 across 22,800 units

Company plan, August 2026

10,200<sup>t</sup>

CO<sub>2</sub>e avoided a year by FY30 – no compressor, no refrigerant; about 0.45 t per device. Cooling already drives 7% of global emissions.

Company model • UNEP, 2023

# From a 50 ml prototype to a deployed platform.

Five years, each version rebuilt from field data — from a bench prototype to ten units with SELCO Foundation to a commercial device. Next: new countries, starting with East Africa and the Gulf.



2021  
**Genesis**  
50 ml solar-powered unit. Proof of concept.



2023  
**Field trials**  
10 units deployed with SELCO Foundation.



2024  
**Version 3.0**  
Redesigned from field data. CDSCO test licence.



2025  
**Commercial-ready**  
Analytics on board. Tested at TÜV SÜD. Patent filed.



2026 →  
**Scale**  
2.5 L, 5 L and 10 L. International expansion.

## Today the device. Tomorrow the chain.

One cold chain, end to end. Samples reach the lab in a 2 L; doses leave the manufacturer in a 10 L, cross the country in a cold container, then step down — a 5 L on a cycle, a 2 L on a health worker's back — to a child under a tree, with one temperature record the whole way.

### THE LONG VIEW

One provider and one temperature record from factory cold room to vaccination session: transport, last-mile devices and site storage, all monitored, all auditable. In this plan: the 2 L today, the 5 L in FY27, the 10 L in FY28. The container and site storage are the ten-to-twenty-year view. The biologics decade needs that chain of custody, and the record that proves it.



# Design, electronics and firmware. All developed in-house.



**Suraj J R**

CO-FOUNDER

Built the cold engine and next-gen controller

**Shruti**

FIRMWARE

Embedded and edge-AI control

**Ankita Mittal**

CEO

ISB • market entry, first institutional deployments

**Prerit Mittal**

FOUNDER

IIM-A • Royal Academy of Engineering Fellow

Chairman • Harkesh Kumar Mittal – architect of India's national incubation programme (NIDHI); chaired DPIIT's Start-up India seed fund committee

Group portrait composed from the team's individual photographs

# \$1.1M at \$8.5M pre-money.

Equity, or a revenue-based structure for funds that prefer it. One round to certification and a thousand units in the field.

|     |                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------|
| 35% | Deployments — India first, then East Africa and Southeast Asia with distributor and NGO partners |
| 25% | Team and field operations — the first hire is a quality manager for ISO 13485                    |
| 25% | 5 L and 10 L manufacturing, the 5 L first                                                        |
| 15% | IP, WHO PQS E003 submission, working capital                                                     |

BEYOND CAPITAL

Introductions to national immunisation programmes, NGO fleets and development finance institutions. We report doses protected, excursions avoided and CO<sub>2</sub>e per unit from live telemetry, so impact measurement costs the fund nothing.

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What this round builds – the 5 L platform, FY27